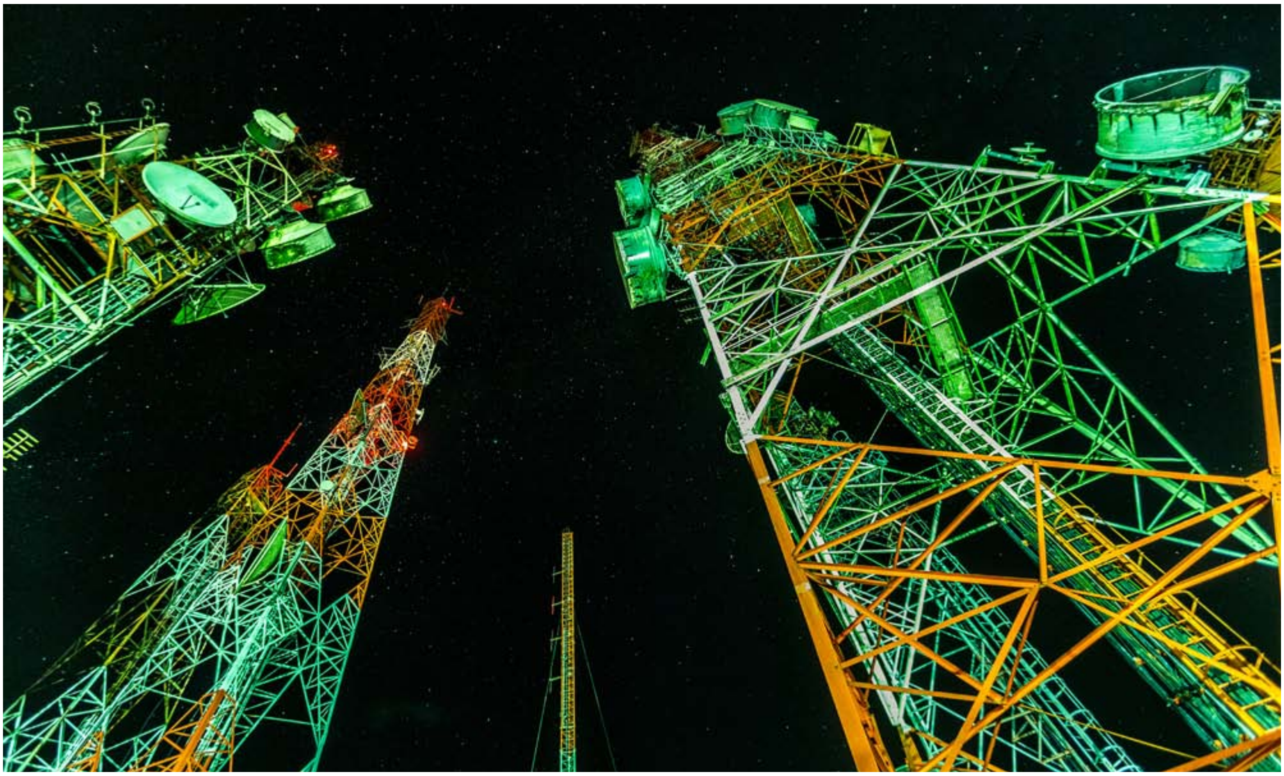
The Role of Network Function Virtualization in Telecom Infrastructure

This sixth article in the series on integration of network function virtualization (NFV) with the DevOps pipeline talks about the advantages of NFV and the best way to manage NFV infrastructure.



Service providers are always looking for solutions that provide the latest telecom applications and comply with the demands of their users. These solutions should not be a bottleneck while scaling or call for hefty financial investment. The hardware deployed is specialised to do a task specific to the service — for example, firewall, monitoring, routing, and more. However, capex and opex investments will see a rise if this proprietary hardware reaches the end of life quickly due to rapid innovations.

Network function virtualization

Network function virtualization (NFV) brings service providers out of this bottleneck. It is a cost-effective implementation of telecommunications infrastructure. The functioning proprietary hardware is replaced with an IT solution with the use of virtualization. NFV connects a wide range of industry-grade servers and hardware, which are driven by a common set of virtual network functions (VNFs). It aims to transform the way network communications connect to network equipment such as switches and storage (the latter is remotely located in either data centres or end user premises).

The book titled 'Network Function Virtualization' by Ken Gray and Tom Nadeau defines NFV as: "NFV